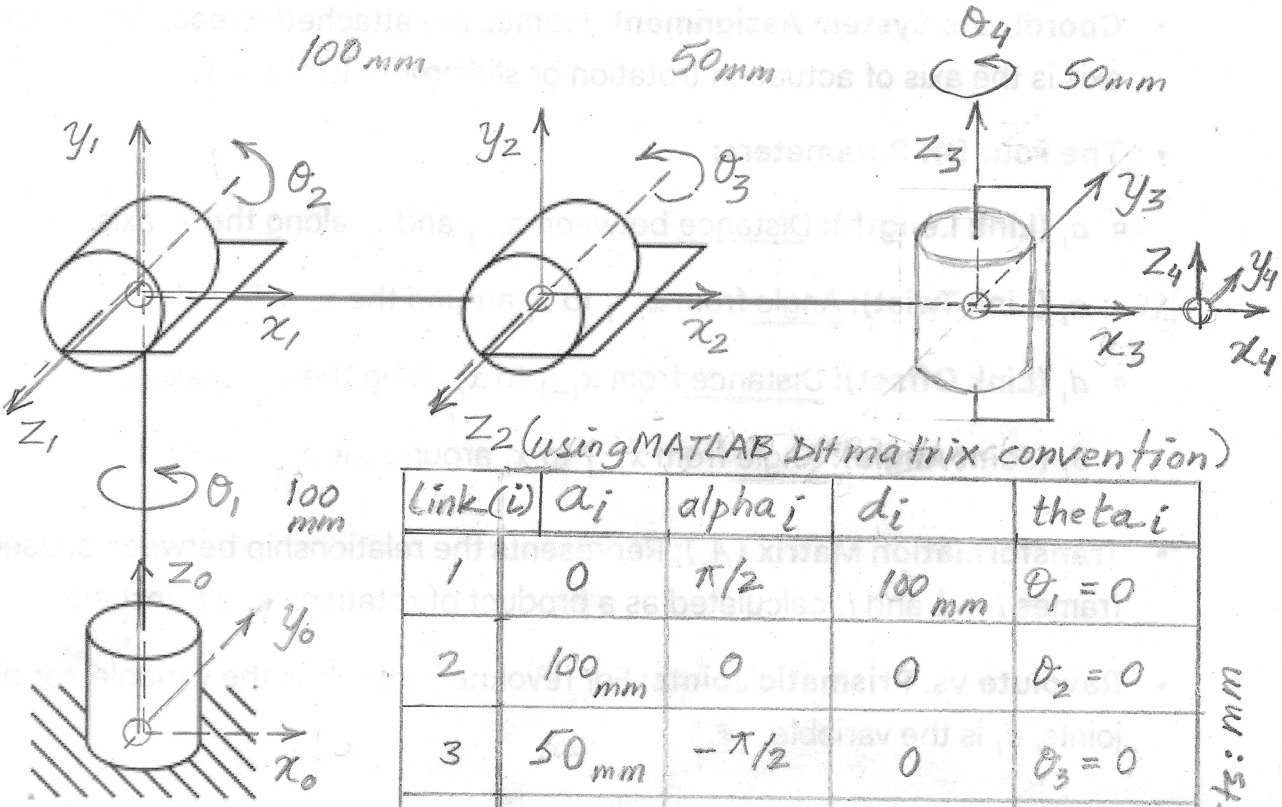
Systematic Procedure (Algorithm)

1. **Identify Axes:** Locate and label the joint axes (z_0 to z_n).
2. **Assign Origins:** Define the origin for each frame.
3. **Define x-axes:** Choose x_i to be perpendicular to both z_{i-1} and z_i.
4. **Define y-axes:** Complete the right-handed frame.
5. **Determine Parameters:** Identify the 4 DH parameters for each link.
6. **Calculate Matrices:** Assemble the A_i transformation matrices.
7. **Final Transformation:** Multiply the matrices ($T = A_1 A_2 ... A_n$) to find the total transformation from base to end-effector.

The convention simplifies the kinematic analysis of n-link manipulators, making it manageable despite high complexity.



Link (i)	a_i	alpha i	d_i	theta i
1	0	$\pi/2$	100 mm	$\theta_1 = 0$
2	100 mm	0	0	$\theta_2 = 0$
3	50 mm	$-\pi/2$	0	$\theta_3 = 0$
4	50 mm	0	0	$\theta_4 = 0$

units: mm

Figure 3.16: Three-link articulated robot.

Four

alpha & theta are angles in radians OR degrees?

initial values ?